

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 6

9 Logic and Proof (lp15)

propositional
logic

- (a) Suppose we have the propositional symbols $P_1, P_2, \dots, P_n$, where $n > 2$, and consider the set of clauses

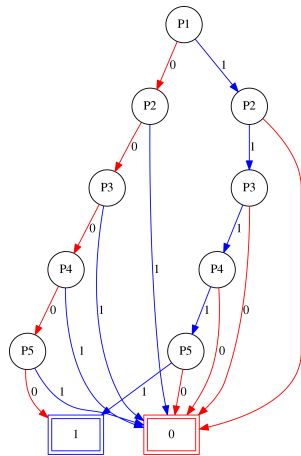
$$\{\neg P_1, P_2\} \quad \{\neg P_2, P_3\} \quad \dots \quad \{\neg P_n, P_1\}.$$

- (i) List the satisfying interpretations (if any) of this set of clauses, with brief justification. [3 marks]

Answer: The clauses form a chain of implications linking back to the start, forcing all the propositional symbols to take on the same value. Therefore there are two satisfying interpretations: with all the symbols set to 1 and with all the symbols set to 0. A good answer should include an argument that those interpretations are the only ones, e.g. by noting that P_1 can only be true or false, or by noting that either $\neg P_n$ or P_1 must be true.

- (ii) Regarding the set of clauses above as a single propositional formula, and using the variable ordering $P_1, P_2, \dots, P_n$, sketch the corresponding BDD. Does the choice of variable ordering here significantly affect the size of the resulting BDD? [5 marks]

Answer: This BDD will have P_1 at the top with a true-link to P_2 , etc., with a chain of true-links continuing to P_n and a further true-link to 1. It will also have a chain of false-links from P_1 at the top continuing to P_n and a further false-link to 1. These represent the two satisfying interpretations. All the other links in the BDD go directly to 0. The total number of decision nodes in the BDD is $2n - 1$. Because all the propositional symbols have to take the same value, the dependencies between them are trivial. We get essentially the same BDD regardless of the variable ordering. Of course, for an arbitrary BDD, the dependence on the ordering can be massive.



- (iii) Briefly describe the set of clauses that would be generated by a resolution theorem prover, starting with the set of clauses above. [3 marks]

Answer: At first there would be n possible pairs of complementary literals, but by symmetry we can confine ourselves to resolving $\{\neg P_1, P_2\}$ with $\{\neg P_2, P_3\}$. The result would be $\{\neg P_1, P_3\}$. Further resolution steps would clearly combine one negative and one positive literal, yielding clauses of the form $\{\neg P_i, P_j\}$ in all combinations. No other clauses would be derived, in particular, not the empty clause.

Clauses like $\{\neg P_1, P_1\}$ are tautologies and get deleted, but this does not halt the process. A good resolution prover will halt once all combinations have been produced.

- (b) For the following set of clauses, either exhibit a model, or show that none exists using resolution. Below, a and b are constants, while y and z are variables.

$$\begin{aligned} &\{P(a, f(a)), P(a, b)\} \\ &\{P(a, f(a)), \neg P(z, b), P(z, f(a))\} \\ &\{\neg P(a, f(a)), P(g(y), y), \neg P(a, y)\} \\ &\{\neg P(a, f(a)), \neg P(g(y), f(a)), \neg P(a, y)\} \end{aligned}$$

[9 marks]

Answer: We immediately see the necessity for factoring because there are no unit clauses. All but the first clause can be factored, and we obtain new clauses:

$$\begin{aligned} &\{P(a, f(a)), \neg P(a, b)\} \\ &\{\neg P(a, f(a)), P(g(f(a), f(a)))\} \\ &\{\neg P(a, f(a)), \neg P(g(f(a), f(a)))\} \end{aligned}$$

Now, resolving on the second literals of the last two clauses, we obtain a unit clause: $\{\neg P(a, f(a))\}$.

Resolving that with the first of the factored clauses above yields another unit clause: $\{\neg P(a, b)\}$.

But these two unit clauses, in two further resolution steps, contradict the first clause. Therefore the clauses are unsatisfiable, and there are no models.
